Supplementary Material

Which reverts count? Reverter populations, attribution rules and observation windows behind a Wikipedia indicator of information quality work

Tables are generated by 03_pipeline/17_supplement_v14.py from the aggregates in 04_output/ of the public repository. Editions marked * were held out for the registered tests. Dump rule: the history dumps' unbounded identity-revert rule. Radius rule: the same hashes with the 15-revision radius of the mwreverts library. Except for Table S7 and the registered rows of Table S8, all analyses are exploratory. Shares based on fewer than 50 edits are not shown; counts are.

S1. Registration, sequence of events and deviations

The registered document (Zenodo, https://doi.org/10.5281/zenodo.23720117; commit 7fcff3d, 28 October 2026, 04:39 UTC) fixed the cohort, the three attribution rules, the reverter filters, the windows, four hypotheses with thresholds, and the code. The sequence was as follows.

1. English, Korean, French and German were analyzed without restriction. The thresholds of H1 to H4 were chosen from these four editions.
2. A batch driver downloaded and processed Japanese, Polish and Italian on 27 and 28 October 2026. The driver wrote its output files to disk and printed progress lines; the summary lines for H1 and H2 were among the lines printed to the terminal log. I did not open the output files or read those lines before the registered commit. This is recorded in the work log and by file and commit timestamps, and it cannot be verified by a reader.
3. The registration was committed and deposited. The results for the held-out editions were committed 11 minutes later, which reflects that they had already been computed.
4. Section 1 of the registered text says that the held-out editions "had not been downloaded". That sentence is wrong (deviation 4 below); the header of the same document states the actual sequence.

The registration is therefore a hold-out replication with thresholds fixed in advance on other editions. It protects against choosing thresholds after seeing the held-out results only to the extent that the reader accepts step 2.

Decisions. H1, H2 and H4 are tested with one-sided 95 percent bootstrap bounds (editors resampled), Holm-adjusted over the three tests within an edition. With three tests the Holm procedure compares the smallest p value with .05/3, the next with .05/2 and the largest with .05. In every supported decision in Table S7 none of the 2,000 bootstrap replicates fell on the wrong side of the threshold. This is a count, not an exact p value: with 2,000 replicates the smallest attainable bootstrap p value is 1/2,001, which is below the strictest Holm level of .05/3, so the adjustment changes no decision. The one unsupported decision (H4, Polish) is unsupported without adjustment. H3 is a descriptive criterion without a test.

Specification of the registered hypotheses (registered text, Section 4; registered code at commit 7fcff3d; outputs 04_output/<wiki>_confirmatory.csv and, for H4 with 2,000 replicates, 04_output/rev_h4_boot2000.csv). Strict: edits first reverted by an excluded class are recoded as not reverted. Replace: the first later revert by a non-excluded reverter is sought.

	Estimand	Data	Variant	Window	Registered criterion	Resampling
H1	D = Δ(24 h) − Δ(unbounded), where Δ is the revert rate with all reverts minus the rate with bots and tools excluded; R = Δ(unbounded)/Δ(24 h) reported	Page-history sample	Replace	24 hours and unbounded	One-sided 95 percent lower bound of D above 0 (R of .75 or more read as most of the effect persisting)	Editors, 2,000
H2	S = 1 − Δ(replace)/Δ(strict), the share of bot- or tool-reverted edits with a later non-bot, non-tool revert	Page-history sample	Replace against strict	24 hours	One-sided 95 percent upper bound of S below .25	Editors, 2,000
H3	Agreement of the tag rule with the radius rule; share of edits reverted under the dump rule and untagged whose first revert lies beyond the radius	Page-history sample, first edits from 1 August 2020 as registered	Neither (detection rules)	Unbounded	Agreement of at least .95; share above .5. Descriptive, no test	None
H4	Ratio of the largest to the smallest odds ratio of retention (reverted against not reverted) across nine cells: all reverts, without bots, without bots and tools, by 24 hours, 7 days and unbounded	Full cohort	Strict	24 hours, 7 days, unbounded	Ratio of at least 1.5 with a 95 percent lower bound above 1.0	Editors; 500 in the registered code, recomputed with 2,000

Holm adjustment is applied over H1, H2 and H4 within an edition.

Deviations (full log: 04_logs/DEVIATION_LOG.md in the repository).

#	Registered	Done	Effect
1	Mixed model as an exploratory complement	Not reported; a per-edition logistic model was run and withdrawn as uninformative	None
2	Robustness list (12 h, 48 h, 30 d windows; five anchors; all registered accounts; historical-only bot flag; undo as a tool; 5-second latency; leave-one-edition-out; by year; Korean former anchor definition)	All run; reported in Table S8. Added: 180-day window, current-only bot flag, exclusion of unregistered reverters, cohort from 2021, Korean surge split	None
3	H3, second clause	Rejected in all three editions and reported as rejected; mechanism examined afterwards (exploratory)	Decision unchanged
4	Text: held-out editions “had not been downloaded”	They had been downloaded and processed; outputs unopened (see above)	Weakens the evidential status of the tests
5	2,000 bootstrap replicates	Registered code used 2,000 for H1 and H2, 500 for H4; H4 recomputed with 2,000	No decision changes
6	Negative control; simulation	Withdrawn: the control cannot fail and the simulation restates a closed form. Outputs remain in the repository	None
7	Interpretation of H2 as human substitution for automation	Withdrawn. Follow-up reverts redefined as reverts by a third party; radius rule, reverter classes, retention by class added (exploratory)	Decisions unchanged; interpretation changed
8	Tag analyses for first edits from August 2020	Coverage analyses restricted to first edits from October 2020 (tag enabled on all wikis 15 September 2020); H3 reported with the registered window	First clause unaffected. Second clause: rejected in all three editions with the registered window; with the October window it would be met in Japanese (Table S10)
9	“Successor filters” F0, F1, F2	Reverter classes; tool exclusion presented as hypothetical	None
10	Release of editor-level derived data with randomized identifiers	Not released: one first edit per editor with its timestamp and edition re-identifies the account from the public history whatever the identifier. Code regenerates the files	Reduces direct reusability; aggregates and code are public
11	Minimum cell size 50	Applied in all tables of the manuscript; an earlier public draft showed some smaller cells	None
12	Holm adjustment “across H1 to H4”	Applied over H1, H2 and H4; H3 has no test statistic. H1: all three R values are at or above the registered .75, read as registered (“most of the filter effect persists”). H4: registered as a point ratio of at least 1.5 with a lower bound above 1.0; reported so	None
13	Model used for coding assistance	The review coding by a language model and its checks are described in the manuscript; an internal pre-submission check of drafts by language-model	None

#	Registered	Done	Effect
		reviewers led to several of the changes above and is recorded in the work log	

Table S1. Survey of reporting practice: how 41 studies using a revert-based measure operationalize it

One row per study. The coding table with the passage supporting each code, URLs, evidence levels, the author's check of each study and the earlier model-to-model comparison is 04_output/practice_review.csv and the files beside it. Search record: 04_output/practice_review_LIS_search.md.

Inclusion. A study is tallied if a revert measure computed from revision histories is an outcome, an exposure, a label or a conflict indicator in its main analysis, or is itself the object of description. Studies in which reverting is one category in a coding of editing activity, or enters only as a control variable, are listed below and not tallied, because radius, window and reverter population are not choices their authors faced in the same form. This criterion was narrowed after a first version of the review had tallied those four studies; the narrowing lowers the counts of unstated choices.

Flow of studies. Stage 1 (citation networks of the methodological works and keyword searches; the number of records screened was not logged): 36 eligible studies, of which 34 were tallied and 2 are listed but not tallied because the full text could not be read. Stage 2a (OpenAlex, information science and neighboring journals): 911 records from six queries and 258 from a title search (1,169 in all); 79 assessed for eligibility; 54 candidates could at first be read only as abstracts or not at all. Stage 2b (12 November 2026, publisher full texts through institutional access and open-access copies): 45 of those 54 candidates were read in full (JASIST 9, Journal of Information Science 3, Information Processing & Management 3, Library & Information Science Research 1, Journal of Documentation 2, Online Information Review 5, New Media & Society 5, Social Science Computer Review 2, Information, Communication & Society 3, Journal of Computer-Mediated Communication 2, Computers in Human Behavior 1, First Monday 5, PLoS ONE 3, Royal Society Open Science 1), and the full text of 242 JASIST and ASIS&T proceedings articles returned by a search for Wikipedia and revert terms was screened by term counts, with every article mentioning Wikipedia at least eight times and reverting at least twice inspected. Retrieval and term-count screening in both stages were run through a language-

model tool and a scripted browser session; eligibility was decided on the retrieved text. Stage 2 yielded 12 studies with a revert-based quantity: 7 are tallied, 4 are listed but not tallied under the inclusion criterion above, and 1 (an earlier conference report of the same sample as Stvilia et al., 2007) is listed but not tallied. The other candidates read contain no revert-based measure or use reverts only as background, as a described example, as one hand-coded category on policy pages without a reported quantity, or as a sample filter. Eleven texts could not be read: the two eligible studies of stage 1 named above, and nine candidates of stage 2 (four PLoS ONE articles and one Physical Review E article that could not be retrieved by the tools used, one New Media & Society article whose file could not be parsed, and three ACM proceedings papers). Total tallied: 41. The codes of one tallied study (Priedhorsky et al., 2007) were proposed from an extract made in the first pass, because no copy could be retrieved for the model's second pass (recorded in notes).

Reading and coding. A language model (Claude, Anthropic; model versions and dates in 04_logs/AI_ASSISTANCE_LOG.md) extracted candidate passages and proposed codes, instructed to code a choice as stated only on finding an establishing passage and to record it. The author checked the codes of every tallied study against the source text and, where available, released code or author copies (column author_check); for four studies, two of them tallied, the check was assisted by a second language model (ChatGPT, OpenAI). Eligibility was proposed by the model on the retrieved text; the author confirmed the list of included studies and did not re-read the candidates the model excluded. The notes and evidence_level columns record where a code rests on an extracted passage and not on a re-read of the full text. No second human coder took part, and no reliability coefficient is claimed for the final codes. For the record of the process: before the coding rules were final, a separate model instance coded a random 13 studies blind and agreed with the first pass on 72 of 78 judgments (kappa 1.00 for the three reverter choices, .78 for detection, .61 for window, .59 for sensitivity analysis); that comparison is between two runs of the same model family on superseded codes and says nothing about the final table. The construct codes were assigned by the author and were not part of it.

Tallies. The article reports one tally (its Table 2): a choice counts as stated only if the article or its supplement states it for the main analysis. A rule stated only for a sub-analysis is not the study's rule. A detection principle named in the text counts as stated even if its implementation is

deferred to a citation (Halfaker et al., 2013); detection given only by citation does not (Lam et al., 2011). The observation window is tallied in the article only for studies whose revert measure is an edit-level outcome, exposure or label, because for conflict measures computed over whole article histories and for descriptive studies it does not apply in the same form. For information, the table below also shows in how many studies an unstated choice can be recovered from a cited procedure, a named library default, the named data source, released code or the design of the measure (the wording “eventually reverted”; a measure built on mutual reverts between two editors). These entries were derived from the notes of the author’s check and confirmed by the author; they are the last column of Table 2 of the article. The whole-history design is marked only where the study states it. Removing bot accounts or bot-authored edits from the sample is not counted as recoverable for reverts made by bots: it concerns who wrote the observed edits, not who reverted them. The ten studies concerned are listed below.

Choice	Stated in the article	Not stated, but recoverable	Neither
Detection rule	29	3	9
Search radius, in the 22 studies using an identity rule	1	4	17
Reverts made by bots	9	0	32
Reverts made with one-click tools	5	0	36
Self-reverts	16	5	20
Observation window, all studies	5	8	28
Observation window, in the 20 studies with an edit-level measure (outcome, exposure or label; Stvilia et al., 2007, whose measures are article-level aggregates, is not among them)	5	6	9

Studies that state nothing on bots, tools or self-reverts: 18 (14 if recoverable choices are counted). Some sensitivity analysis over a revert choice: 8; over the reverter population: 0.

By construct (constructs assigned by the author; four studies that evaluate detection methods or report descriptive rates are under other).

	Studies	No statement on bot, tool or self-reverts	No window stated
quality judgment	11	6	8
conflict	15	6	15
rejection	9	3	7
effort	2	0	2
other	4	3	4
all	41	18	36

Codes.

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
Kittur et al., 2007	CHI 2007	en only	conflict	identity + summary matching	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Priedhorsky et al., 2007	GROUP '07	en only	quality judgment	identity + summary matching	15 revisions	counted explicitly	counted explicitly	not stated	not stated	partial	citation-network search
Halfaker et al., 2009	WikiSym '09	en only	rejection	identity principle (implementation not reported)	none stated	not stated	not stated	excluded	not stated	yes: vandalism reverts removed	citation-network search
Suh et al., 2009	WikiSym '09	en only	rejection	summary matching only	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Ekstrand & Riedl, 2009	WikiSym '09	en only	other	identity hash	none stated	not stated	not stated	not stated	not stated	no	citation-network search
West et al., 2010	EUROSEC '10	en only	quality judgment	rollback or undo actions only	none stated	excluded	counted explicitly	not stated	not stated	no	citation-network search
Halfaker et al., 2011a	WikiSym '11	en only	rejection	not stated	none stated	not stated	not stated	excluded	not stated	no	citation-network search
Halfaker et al., 2011b	WikiSym '11	en only	rejection	rollback or undo actions only	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Lam et al., 2011	WikiSym '11	en only	rejection	not stated	none stated	not stated	not stated	not stated	time-bounded: 7 days	no	citation-network search
Sumi et al., 2011	IEEE PASSAT/SocialCom 2011	multilingual	conflict	identity principle (implementation not reported)	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Yasseri et al., 2012	PLoS ONE	en only	conflict	identity hash	none stated	not stated	not stated	excluded	not stated	no	citation-network search
Flöck et al., 2012	HT '12	en only	other	diff-based	20 revisions (also 50, 100)	not stated	not stated	not stated	not stated	yes: scan range 20/50/100; diff vs identity	citation-network search
Halfaker et al., 2013	American Behavioral Scientist	en only	rejection	identity principle (implementation not reported)	none stated	modelled separately or focal	modelled separately	excluded (definitional)	not stated	no	citation-network search
Geiger & Halfaker, 2013	WikiSym '13	en only	effort	not stated	none stated	modelled separately or focal	modelled separately	not stated	not stated	no	citation-network search
Yasseri et al., 2014	Global Wikipedia (edited book)	multilingual	conflict	identity hash	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Tsvetkova et al., 2016	Scientific Reports	multilingual	conflict	identity hash	none stated	excluded	not stated	excluded	not stated	no	citation-network search
Tsvetkova et al., 2017	PLoS ONE	multilingual	conflict	identity hash	none stated	modelled separately or focal	not stated	excluded	not stated	no	citation-network search
Geiger & Halfaker, 2017	PACMHCI (CSCW)	multilingual	conflict	identity hash	mwreverts default (not reported)	modelled separately or focal	not stated	not stated	not stated	partial	citation-network search

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
TeBlunthuis et al., 2018	CHI 2018	Wikia	rejection	not stated	none stated	modelled separately or focal	not stated	not stated	not stated	no	citation-network search
DeDeo, 2013	PLoS ONE	en only	conflict	identity hash	none stated	not stated	not stated	excluded	not stated	partial	citation-network search
Zhang et al., 2019	ICWSM 2019	en only	rejection	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Tran et al., 2020	IEEE S&P 2020	en only	quality judgment	identity hash	mwreverts default (not reported)	not stated	not stated	excluded	not stated	partial	citation-network search
Hill & Shaw, 2021	Communication Research	Wikia	quality judgment	identity hash	mwreverts default (not reported)	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Chhabra et al., 2020	OpenSym 2020	en only	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	citation-network search
TeBlunthuis et al., 2021	PACM HCI (CSCW)	multilingual	rejection	identity hash	none stated	not stated	not stated	excluded	time-bounded: 48 hours	no	citation-network search
Ruprechter et al., 2021	Scientific Reports	multilingual	other	dump precomputed field	none stated	counted explicitly	not stated	not stated	not stated	no	citation-network search
Wikimedia Foundation, 2021	Meta-Wiki report	single non-English edition (pt)	quality judgment	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	citation-network search
Tran et al., 2022	PACM HCI (CSCW)	multilingual	quality judgment	identity principle (implementation not reported)	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Wang & Zhu, 2022	FAccT '22	multilingual	effort	identity principle (implementation not reported)	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Champion & Hill, 2023	PACM HCI (CSCW)	en only	quality judgment	identity principle (implementation not reported)	none stated	not stated	not stated	not stated	revision-bounded: 10 contributions	no	citation-network search
Warncke-Wang et al., 2023	PACM HCI (CSCW)	multilingual	quality judgment	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	citation-network search
Trokhymovych et al., 2023	KDD '23	multilingual	quality judgment	not stated	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Franzmeyer et al., 2024	Findings of ACL 2024	en only	quality judgment	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	citation-network search
Formisano et al., 2026	Humanities and Social Sciences Communications	en only	other	dump precomputed field	none stated	not stated	not stated	not stated	not stated	no	citation-network search
Stvilia et al., 2007	Journal of the American Society	en only	quality judgment	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal

short_cite	venue	editions	construct	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	source
	for Information Science and Technology										search
Iñiguez et al., 2014	EPJ Data Science	multilingual	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Rudas et al., 2017	PLoS ONE	multilingual	conflict	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Yasseri & Kertész, 2013	Journal of Statistical Physics	multilingual	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Sepehri Rad & Barbosa, 2012	WikiSym '12	not stated	conflict	not stated	none stated	not stated	not stated	not stated	not stated	no	LIS and neighbouring journal search
Morris-O'Connor et al., 2023	Journal of Documentation	en only	conflict	identity hash	none stated	not stated	not stated	not stated	not stated	no	LIS full-text screening (library access)
Ariel, 2026	Journal of Documentation	single non-English edition (he)	conflict	change tags and summary matching	none stated	not stated	counted explicitly	not stated	not stated	no	LIS full-text screening (library access)

Studies that remove bots from the sample without stating how reverts made by bots are treated. Refers to: authors (the accounts whose edits are observed) or unclear.

Study	Exclusion refers to	Basis in the coding notes
Halfaker et al., 2009	authors	Footnote to the sample description: revisions attributed to registered editors, with bots removed.
Suh et al., 2009	authors	Bots are defined as edits made by a user whose user id is registered as a bot; nothing on reverts performed by bots.
Zhang et al., 2019	authors	A footnote excluding bots is attached to the definition of the editor crowd.
TeBlunthuis et al., 2021	authors	Bot-authored revisions are excluded; the paper notes that bots revert edits and states no rule for them.
Wikimedia Foundation, 2021	authors	Bot edits are removed from one metric (net non-reverted edits); nothing on reverts performed by bots.
Wang & Zhu, 2022	authors	Revisions by robots are removed from the dataset.
Champion & Hill, 2023	authors	Section 4.2 (data and measures): revisions made by bots are omitted, identified from a list of registered bots and a historical dataset; nothing on reverts performed by bots (arXiv:2308.06403v2, read 12 November 2026).
Trokhymovych et al., 2023	authors	Section 4.2.2 (user type filter): bot edits are removed from the dataset; nothing on reverts

Study	Exclusion refers to	Basis in the coding notes
		performed by bots (arXiv:2306.01650, read 12 November 2026).
Rudas et al., 2017	unclear	Bot-flagged users are excluded from editor activity statistics; the treatment of reverting accounts is not described.
Morris-O'Connor et al., 2023	unclear	Bot accounts are disregarded in editing profiles; whether reverts by bots are removed from the revert graph is not established.

Choices recoverable from outside the article's own statement.

short_cite	detection	radius	bot	self	window
Halfaker et al., 2009					"eventually reverted" read as unbounded
Lam et al., 2011	cited procedure (Priedhorsky et al., 2007)				
Geiger & Halfaker, 2013					"eventually reverted" read as unbounded
Geiger & Halfaker, 2017		library default (mwreverts, 15); released code sets 15		released code drops self-reverts	
TeBlunthuis et al., 2018	cited study followed (Halfaker et al., 2013)				
Zhang et al., 2019					"eventually reverted" read as unbounded
Tran et al., 2020		library default (mwreverts, 15)			
Hill & Shaw, 2021		library default (mwreverts, 15)			
Ruprechter et al., 2021					dump field (unbounded to the dump date)
Tran et al., 2022					"eventually reverted" read as unbounded
Trokhymovych et al., 2023	dump field visible in a data-sample figure				dump field (unbounded to the dump date)
Formisano et al., 2026					dump field (unbounded to the dump date)
Iñiguez et al., 2014		all pairs of revisions over the whole history (unbounded)		measure built on mutual reverts between two editors	whole article history
Yasseri & Kertész, 2013				measure built on mutual reverts between two editors	
Sepehri Rad & Barbosa, 2012				measure built on mutual reverts between two editors	
Morris-O'Connor et al., 2023				measure built on mutual reverts between two editors	

Listed, not tallied.

short_cite	venue	reason
Kiesel et al., 2017	ICWSM 2017	full text could not be read
Boté-Vericad, 2026	Zenodo report	full text could not be read
Stvilia et al., 2005	ICIQ 2005 (International Conference on Information Quality)	same sample and measures as Stvilia et al. (2007)
Purdy, 2020	First Monday	revert is one hand-coded activity category

short_cite	venue	reason
Lerner & Lomi, 2018	PLoS ONE	revert count enters only as a control variable
Arazy et al., 2019	Journal of the Association for Information Science and Technology	revert is one annotated activity category
Gao et al., 2026	Journal of the Association for Information Science and Technology	revert is one classifier-assigned action type

Reporting items 1 to 4 (Table 4 of the article) applied to the reviewed studies. Item 1 (construct) could be assigned by the author for all studies but is seldom named by the study itself; item 2 (attribution rule) is stated in 29 of 41; item 3 (window) in 5; item 4 (bots, tools and self-reverts, all three) in 1, at least one of the three in 23.

References for Table S1

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Table S2. Composition of first reverts of newcomer first edits (dump rule, full cohorts)

First edits 2019 onward, window 24 h. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.37	0.122 [0.001, 0.300]	0.448 [0.346, 0.534]	0.364 [0.286, 0.432]	0.023 [0.018, 0.027]	0.044 [0.034, 0.053]
Italian*	135,400	0.342	0.000 [0.000, 0.001]	0.525 [0.455, 0.583]	0.407 [0.354, 0.468]	0.024 [0.019, 0.032]	0.044 [0.035, 0.058]
Korean	47,633	0.216	0.003 [0.000, 0.008]	0.516 [0.349, 0.618]	0.289 [0.232, 0.370]	0.096 [0.068, 0.147]	0.096 [0.068, 0.146]
German	236,903	0.256	0.001 [0.000, 0.003]	0.410 [0.351, 0.466]	0.545 [0.492, 0.599]	0.017 [0.014, 0.020]	0.027 [0.023, 0.033]
Polish*	67,755	0.222	0.001 [0.000, 0.004]	0.315 [0.263, 0.374]	0.616 [0.556, 0.668]	0.030 [0.023, 0.041]	0.038 [0.029, 0.050]
French	346,938	0.243	0.042 [0.000, 0.125]	0.255 [0.189, 0.324]	0.643 [0.564, 0.719]	0.012 [0.010, 0.015]	0.047 [0.039, 0.058]
Japanese*	223,045	0.14	0.001 [0.000, 0.002]	0.247 [0.146, 0.339]	0.558 [0.491, 0.631]	0.080 [0.066, 0.098]	0.114 [0.094, 0.140]

First edits 2019 onward, window unbounded. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.412	0.111 [0.001, 0.274]	0.432 [0.341, 0.508]	0.383 [0.308, 0.448]	0.034 [0.027, 0.040]	0.040 [0.032, 0.048]
Italian*	135,400	0.369	0.001 [0.000, 0.002]	0.503 [0.428, 0.561]	0.421 [0.369, 0.482]	0.034 [0.027, 0.044]	0.041 [0.033, 0.053]
Korean	47,633	0.279	0.006 [0.000, 0.015]	0.438 [0.286, 0.544]	0.336 [0.282, 0.405]	0.142 [0.103, 0.206]	0.078 [0.055, 0.114]
German	236,903	0.294	0.003 [0.000,	0.369 [0.314,	0.586 [0.533,	0.017 [0.014,	0.025 [0.021,

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
			0.007]	0.427]	0.637]	0.021]	0.030]
Polish*	67,755	0.303	0.002 [0.000, 0.005]	0.248 [0.197, 0.306]	0.690 [0.627, 0.745]	0.031 [0.024, 0.041]	0.029 [0.022, 0.038]
French	346,938	0.27	0.039 [0.001, 0.116]	0.241 [0.177, 0.305]	0.654 [0.583, 0.725]	0.023 [0.019, 0.027]	0.044 [0.036, 0.053]
Japanese*	223,045	0.172	0.003 [0.000, 0.007]	0.209 [0.124, 0.295]	0.581 [0.518, 0.643]	0.109 [0.091, 0.130]	0.097 [0.081, 0.115]

First edits 2021 onward, window 24 h. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.361	0.125 [0.001, 0.304]	0.524 [0.407, 0.620]	0.285 [0.223, 0.346]	0.022 [0.017, 0.027]	0.044 [0.034, 0.054]
Italian*	88,005	0.334	0.000 [0.000, 0.000]	0.546 [0.467, 0.611]	0.390 [0.331, 0.457]	0.020 [0.015, 0.028]	0.044 [0.033, 0.060]
Korean	32,751	0.239	0.001 [0.000, 0.003]	0.554 [0.389, 0.648]	0.262 [0.206, 0.345]	0.094 [0.065, 0.148]	0.090 [0.062, 0.140]
German	159,216	0.248	0.001 [0.000, 0.003]	0.394 [0.332, 0.456]	0.560 [0.500, 0.617]	0.018 [0.015, 0.022]	0.028 [0.023, 0.034]
Polish*	45,382	0.227	0.002 [0.000, 0.005]	0.303 [0.244, 0.368]	0.624 [0.562, 0.679]	0.035 [0.025, 0.048]	0.037 [0.027, 0.049]
French	229,556	0.232	0.020 [0.000, 0.062]	0.261 [0.194, 0.325]	0.656 [0.595, 0.721]	0.011 [0.009, 0.013]	0.051 [0.042, 0.062]
Japanese*	148,508	0.138	0.001 [0.000, 0.003]	0.300 [0.171, 0.411]	0.519 [0.441, 0.612]	0.069 [0.054, 0.089]	0.110 [0.086, 0.141]

First edits 2021 onward, window unbounded. Shares of first reverts; intervals are 95 percent percentile intervals from 2,000 bootstrap resamples of reverting accounts (each revert by an unregistered editor counted as its own account).

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.402	0.114 [0.001, 0.280]	0.508 [0.401, 0.594]	0.305 [0.243, 0.364]	0.033 [0.026, 0.039]	0.040 [0.032, 0.049]
Italian*	88,005	0.361	0.001 [0.000, 0.001]	0.522 [0.445, 0.583]	0.406 [0.351, 0.471]	0.030 [0.023, 0.040]	0.041 [0.032, 0.056]
Korean	32,751	0.303	0.004 [0.000, 0.009]	0.477 [0.307, 0.581]	0.310 [0.256, 0.386]	0.135 [0.094, 0.207]	0.074 [0.052, 0.115]
German	159,216	0.287	0.003 [0.000, 0.007]	0.353 [0.297, 0.415]	0.600 [0.542, 0.653]	0.018 [0.015, 0.022]	0.025 [0.021, 0.030]
Polish*	45,382	0.316	0.002 [0.000, 0.005]	0.234 [0.174, 0.299]	0.704 [0.636, 0.767]	0.033 [0.024, 0.045]	0.027 [0.020, 0.037]
French	229,556	0.26	0.019 [0.001, 0.059]	0.246 [0.184, 0.307]	0.667 [0.607, 0.728]	0.020 [0.017, 0.025]	0.047 [0.040, 0.057]
Japanese*	148,508	0.17	0.004 [0.000, 0.010]	0.254 [0.145, 0.361]	0.553 [0.477, 0.634]	0.096 [0.077, 0.119]	0.093 [0.074, 0.115]

Concentration of reverting, first edits 2019 onward, 24 hours.

Edition	Bot: accounts	Bot: share of the most active account	One-click tool: accounts	One-click tool: share of the most active account	Other registered, manual: accounts	Other registered, manual: share of the most active account
English	29	0.931	6,596	0.061	22,356	0.017
Italian*	6	0.417	201	0.064	1,100	0.068
Korean	5	0.484	93	0.187	521	0.09
German	6	0.395	1,286	0.09	2,615	0.053
Polish*	2	0.789	341	0.109	726	0.101
French	10	0.981	316	0.125	2,756	0.039
Japanese*	5	0.714	103	0.234	2,285	0.045

First edits that create a page (page-history samples). Such edits cannot be identity-reverted; removal is by deletion, which is outside the indicator. English restricts article creation to autoconfirmed accounts.

Edition	Share of first edits that create a page	p reverted within 24 h, all	p reverted within 24 h, edits to existing pages	First reverts made by the revision that follows the measured edit
English	0	0.367	0.367	0.731
Italian*	0.155	0.341	0.403	0.72
Korean	0.145	0.216	0.252	0.643
German	0.16	0.256	0.304	0.788
Polish*	0.201	0.221	0.277	0.719
French	0.121	0.243	0.277	0.721
Japanese*	0.093	0.14	0.155	0.672

Table S3. Change in the revert rate when a class of first reverts is recoded as not reverted, and what the tool class can see

Percentage points; by identity, the class's share of first reverts times the rate.

First edits 2019 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts by other registered editors
English	4.53	16.56	1.62	0.84	11.24	0.784
Italian*	0.01	17.93	1.49	0.83	10.17	0.699
Korean	0.07	11.13	2.07	2.07	5.87	0.71
German	0.04	10.5	0.69	0.43	7.31	0.489
Polish*	0.03	6.99	0.83	0.67	7.54	0.502
French	1.02	6.2	1.15	0.3	11.69	0.722
Japanese*	0.01	3.45	1.6	1.11	5.69	0.618

First edits 2021 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts by other registered editors
English	4.51	18.9	1.58	0.79	8.49	0.755
Italian*	0	18.21	1.46	0.67	9.76	0.725
Korean	0.03	13.25	2.15	2.24	5.97	0.71
German	0.03	9.76	0.68	0.45	7.19	0.48
Polish*	0.04	6.87	0.83	0.79	8.37	0.532

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts by other registered editors
French	0.48	6.07	1.19	0.26	11.81	0.749
Japanese*	0.02	4.15	1.52	0.96	5.46	0.645

Wider tool tags. Share of first reverts within 24 hours carrying a tag of RedWarn, Ultraviolet, AntiVandal script, STiki, LiveRC or MassRollback but outside the tool class used in the article (rollback, Huggle, Twinkle). Adding them would raise the tool share by the amount shown.

Edition	First edits	First reverts within 24 h	Tool class as used	Wider tags, outside the tool class
English	2019+	809720	0.448	0.039
English	2021+	524398	0.524	0.049
Italian*	2019+	46260	0.525	0.000
Italian*	2021+	29357	0.546	0.000
Korean	2019+	10279	0.516	0.000
Korean	2021+	7838	0.554	0.000
German	2019+	60665	0.410	0.000
German	2021+	39495	0.394	0.000
Polish*	2019+	15033	0.315	0.000
Polish*	2021+	10281	0.303	0.000
French	2019+	84351	0.255	0.085
French	2021+	53369	0.261	0.080
Japanese*	2019+	31127	0.247	0.000
Japanese*	2021+	20515	0.300	0.000

Tool-related change tags on reverting revisions in the article namespace, whole period, by edition (tags matching rollback, Huggle, Twinkle, RedWarn, Ultraviolet, STiki, AntiVandal, SWViewer, LiveRC and similar names; tags not listed for an edition do not occur there).

Edition	Tag	Reverting revisions
Korean	mw-rollback	218443
Korean	app-rollback	23
Japanese*	mw-rollback	160250
Japanese*	MassRollback	19619
Japanese*	app-rollback	11
Polish*	mw-rollback	290595
Polish*	app-rollback	109
Italian*	mw-rollback	1082706
Italian*	app-rollback	240
German	mw-rollback	1139847
German	app-rollback	400
French	mw-rollback	587008
French	LiveRC	285273
French	huggle	28943
French	app-rollback	68
English	mw-rollback	6999688
English	twinkle	3271154
English	huggle	1210355

Edition	Tag	Reverting revisions
English	RedWarn	433575
English	Ultraviolet	157322
English	AntiVandal script	156948
English	STiki	97797
English	app-rollback	1240

Table S4. Follow-up manual reverts by a third party, and the window ratio R, under two attribution rules (page-history samples)

Share of first edits first reverted by the class shown that were later reverted manually (no bot flag, no tool tag) by a registered account that is neither the author nor the account that made the first revert. The window is counted from the first revert. Unregistered follow-up reverters are not counted, because they cannot be compared with the first reverter; tables that count them are marked.

Dump rule, 24 h.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.077	0.079	0.069	0.057	0.060	0.090	0.097
Manual, not the author	0.066	0.084	0.064	0.041	0.037	0.070	0.092
Bot	0.126		0.078	0.063		0.217	
One-click tool	0.064	0.079	0.069	0.057	0.060	0.070	0.098
Other registered, manual	0.066	0.082	0.068	0.041	0.035	0.070	0.091
Unregistered	0.057	0.117	0.056	0.070	0.077	0.061	0.094
Self	0.079	0.084	0.155	0.099	0.100	0.073	0.076

Dump rule, 7 d.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.117	0.133	0.135	0.084	0.098	0.138	0.128
Manual, not the author	0.110	0.133	0.115	0.064	0.065	0.111	0.126
Bot	0.173		0.169	0.063		0.271	
One-click tool	0.103	0.133	0.135	0.084	0.099	0.116	0.129
Other registered, manual	0.111	0.131	0.119	0.063	0.062	0.111	0.125
Unregistered	0.096	0.156	0.103	0.085	0.119	0.097	0.136
Self	0.117	0.143	0.206	0.118	0.125	0.098	0.092

Dump rule, unbounded.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.182	0.229	0.272	0.160	0.220	0.241	0.203
Manual, not the author	0.174	0.211	0.212	0.125	0.154	0.196	0.187
Bot	0.243		0.273	0.159		0.363	
One-click tool	0.166	0.230	0.272	0.160	0.221	0.222	0.204
Other registered, manual	0.176	0.209	0.218	0.124	0.152	0.197	0.185
Unregistered	0.151	0.236	0.200	0.170	0.200	0.174	0.195
Self	0.165	0.218	0.286	0.202	0.204	0.175	0.132

Dump rule, unbounded: follow-up by any third party, registered or not, after the first edit's content had been reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.014	0.021	0.122	0.017	0.028	0.010	0.042
Manual, not the author	0.016	0.021	0.070	0.016	0.014	0.013	0.042

Dump rule, unbounded: follow-up by any third party, registered or not, immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.011	0.015	0.024	0.010	0.016	0.008	0.007
Manual, not the author	0.011	0.015	0.024	0.004	0.007	0.008	0.014

Dump rule, unbounded: follow-up by any third party, registered or not, after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.176	0.217	0.208	0.145	0.191	0.234	0.177
Manual, not the author	0.175	0.204	0.240	0.114	0.146	0.188	0.178

Radius rule, 24 h.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.010	0.013	0.017	0.009	0.013	0.013	0.029
Manual, not the author	0.006	0.011	0.012	0.003	0.003	0.005	0.012
Bot	0.017		0.026	0.032		0.044	
One-click tool	0.008	0.013	0.016	0.009	0.013	0.008	0.029
Other registered, manual	0.006	0.010	0.010	0.003	0.002	0.005	0.013
Unregistered	0.011	0.030	0.015	0.004	0.025	0.006	0.009
Self	0.019	0.029	0.064	0.052	0.058	0.024	0.032

Radius rule, 7 d.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.013	0.016	0.023	0.011	0.017	0.016	0.031
Manual, not the author	0.009	0.013	0.016	0.003	0.004	0.006	0.017
Bot	0.020		0.039	0.032		0.048	
One-click tool	0.011	0.016	0.023	0.010	0.017	0.010	0.032
Other registered, manual	0.008	0.011	0.013	0.003	0.003	0.006	0.017
Unregistered	0.014	0.031	0.022	0.004	0.032	0.006	0.015
Self	0.026	0.035	0.072	0.057	0.060	0.029	0.035

Radius rule, unbounded.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.017	0.019	0.032	0.012	0.023	0.019	0.035
Manual, not the author	0.011	0.016	0.022	0.004	0.006	0.008	0.021
Bot	0.026		0.053	0.048		0.052	
One-click tool	0.014	0.019	0.032	0.012	0.023	0.014	0.035
Other registered, manual	0.011	0.014	0.019	0.004	0.005	0.008	0.022

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Unregistered	0.014	0.036	0.029	0.004	0.034	0.008	0.019
Self	0.031	0.039	0.077	0.064	0.066	0.031	0.038

Radius rule, unbounded: follow-up by any third party, registered or not, after the first edit's content had been reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.001	0.001	0.009	0.001	0.002	0.000	0.004
Manual, not the author	0.001	0.002	0.007	0.000	0.001	0.000	0.002

Radius rule, unbounded: follow-up by any third party, registered or not, immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.010	0.014	0.020	0.010	0.016	0.008	0.007
Manual, not the author	0.011	0.014	0.018	0.004	0.007	0.008	0.013

Radius rule, unbounded: follow-up by any third party, registered or not, after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.011	0.012	0.023	0.005	0.010	0.013	0.029
Manual, not the author	0.006	0.011	0.016	0.002	0.004	0.005	0.015

Dump rule, 24 hours, by the number of revisions on the page before the measured edit (fixed before the first revert).

key	English	Italian*	Korean	German	Polish*	French	Japanese*
1. under 50, Bot or one-click tool	0.056	0.045	0.046	0.045	0.049	0.057	0.069
1. under 50, Other registered, manual	0.039	0.053	0.047	0.036	0.023	0.039	0.061
2. 50 to 499, Bot or one-click tool	0.078	0.071	0.066	0.061	0.059	0.082	0.097
2. 50 to 499, Other registered, manual	0.075	0.074	0.068	0.037	0.035	0.065	0.091
3. 500 or more, Bot or one-click tool	0.103	0.101	0.113	0.061	0.074	0.115	0.126
3. 500 or more, Other registered, manual	0.092	0.101	0.092	0.056	0.061	0.092	0.120

Radius rule, 24 hours, by the number of revisions on the page before the measured edit (fixed before the first revert).

key	English	Italian*	Korean	German	Polish*	French	Japanese*
1. under 50, Bot or one-click tool	0.007	0.007	0.008	0.008	0.010	0.009	0.014
1. under 50, Other registered, manual	0.002	0.007	0.004	0.003	0.000	0.002	0.006
2. 50 to 499, Bot or one-click tool	0.010	0.011	0.013	0.009	0.012	0.013	0.035
2. 50 to 499, Other registered,	0.007	0.007	0.012	0.003	0.002	0.005	0.016

key	English	Italian*	Korean	German	Polish*	French	Japanese*
manual							
3. 500 or more, Bot or one-click tool	0.016	0.017	0.038	0.009	0.018	0.014	0.031
3. 500 or more, Other registered, manual	0.009	0.014	0.014	0.003	0.005	0.007	0.013

Dump rule, 24 hours counted from the measured edit (the registered clock; follow-up by any third party, registered or not).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.086	0.088	0.090	0.063	0.069	0.096	0.106
Manual, not the author	0.087	0.103	0.140	0.053	0.056	0.078	0.142
Bot	0.135					0.227	
One-click tool	0.073	0.088	0.089	0.063	0.069	0.074	0.107
Other registered, manual	0.080	0.095	0.113	0.050	0.052	0.076	0.126
Unregistered	0.200	0.237	0.222	0.131	0.156	0.175	0.259
Self	0.092	0.104	0.197	0.117	0.115	0.082	0.084

Window ratio R (replace effect of excluding bots and tools at the longer window divided by the effect within 24 hours; 2,000 bootstrap resamples of editors). With N(W) the number of first edits first reverted by a bot or tool within W and S(W) the share of them with a later non-bot, non-tool revert within W, the replace effect is N(W)(1 − S(W))/n, so R = N(∞)(1 − S(∞)) / [N(24 h)(1 − S(24 h))]. R falls below 1 when follow-up accumulates faster than first reverts by bots and tools, and rises above 1 when it does not.

key	English	Italian*	Korean	German	Polish*	French	Japanese*
Dump rule, 180 d / 24 h	0.926 [0.919, 0.934]	0.857 [0.848, 0.865]	0.809 [0.793, 0.825]	0.931 [0.921, 0.941]	0.905 [0.890, 0.921]	0.875 [0.861, 0.889]	0.926 [0.906, 0.945]
Dump rule, unbounded / 24 h	0.922 [0.915, 0.930]	0.843 [0.834, 0.852]	0.775 [0.758, 0.792]	0.917 [0.907, 0.928]	0.877 [0.860, 0.892]	0.858 [0.844, 0.871]	0.919 [0.898, 0.939]
Radius rule, 180 d / 24 h	1.043 [1.038, 1.047]	1.024 [1.019, 1.028]	1.074 [1.064, 1.085]	1.036 [1.030, 1.042]	1.067 [1.057, 1.077]	1.039 [1.032, 1.046]	1.046 [1.034, 1.058]
Radius rule, unbounded / 24 h	1.045 [1.041, 1.050]	1.024 [1.020, 1.028]	1.074 [1.064, 1.085]	1.036 [1.031, 1.042]	1.067 [1.057, 1.077]	1.039 [1.032, 1.046]	1.048 [1.037, 1.060]

Table S5. Retention (an edit on days 2 to 60) by who made the first revert within 24 hours

Cells: share retained; odds ratio against newcomers not reverted within 24 hours with 95 percent interval; number of newcomers.

All newcomers.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Any revert	0.097; OR 0.43 [0.43, 0.43]; n = 809,720	0.090; OR 0.40 [0.38, 0.41]; n = 46,260	0.079; OR 0.32 [0.29, 0.34]; n = 10,279	0.107; OR 0.51 [0.50, 0.53]; n = 60,665	0.113; OR 0.54 [0.51, 0.57]; n = 15,033	0.089; OR 0.47 [0.46, 0.49]; n = 84,351	0.116; OR 0.36 [0.35, 0.37]; n = 31,127
Bot	0.069; OR 0.30 [0.29,			0.081; OR 0.40 [0.19,		0.041; OR 0.21 [0.18,	

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
	0.31]; n = 99,135			0.85]; n = 86		0.25]; n = 3,552	
One-click tool	0.086; OR 0.38 [0.37, 0.38]; n = 362,541	0.072; OR 0.31 [0.29, 0.33]; n = 24,278	0.035; OR 0.13 [0.12, 0.16]; n = 5,301	0.062; OR 0.29 [0.27, 0.30]; n = 24,876	0.067; OR 0.30 [0.27, 0.34]; n = 4,733	0.052; OR 0.27 [0.25, 0.28]; n = 21,513	0.037; OR 0.10 [0.09, 0.12]; n = 7,690
Other registered, manual	0.114; OR 0.52 [0.51, 0.52]; n = 294,376	0.108; OR 0.48 [0.46, 0.51]; n = 18,823	0.118; OR 0.50 [0.44, 0.56]; n = 2,971	0.142; OR 0.71 [0.68, 0.73]; n = 33,055	0.135; OR 0.66 [0.62, 0.70]; n = 9,263	0.104; OR 0.57 [0.55, 0.58]; n = 54,268	0.129; OR 0.41 [0.39, 0.43]; n = 17,362
Unregistered	0.106; OR 0.48 [0.45, 0.50]; n = 18,309	0.121; OR 0.55 [0.46, 0.66]; n = 1,130	0.122; OR 0.52 [0.43, 0.63]; n = 988	0.084; OR 0.40 [0.32, 0.50]; n = 1,007	0.132; OR 0.65 [0.50, 0.85]; n = 454	0.082; OR 0.44 [0.35, 0.55]; n = 1,044	0.152; OR 0.49 [0.44, 0.55]; n = 2,485
Self	0.139; OR 0.65 [0.63, 0.67]; n = 35,359	0.128; OR 0.59 [0.51, 0.67]; n = 2,017	0.153; OR 0.67 [0.56, 0.79]; n = 988	0.113; OR 0.55 [0.47, 0.64]; n = 1,641	0.124; OR 0.61 [0.47, 0.78]; n = 564	0.119; OR 0.66 [0.60, 0.73]; n = 3,974	0.199; OR 0.68 [0.63, 0.74]; n = 3,562

Share retained among newcomers not reverted within 24 hours: English 0.199; Italian* 0.201; Korean 0.213; German 0.189; Polish* 0.191; French 0.171; Japanese* 0.266.

Newcomers not blocked within 60 days of the first edit.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Any revert	0.096; OR 0.43 [0.43, 0.43]; n = 785,031	0.088; OR 0.40 [0.38, 0.41]; n = 41,280	0.080; OR 0.32 [0.30, 0.35]; n = 9,508	0.110; OR 0.53 [0.51, 0.54]; n = 57,461	0.112; OR 0.54 [0.51, 0.57]; n = 14,302	0.090; OR 0.48 [0.47, 0.50]; n = 78,726	0.112; OR 0.35 [0.34, 0.36]; n = 28,635
Bot	0.069; OR 0.30 [0.29, 0.31]; n = 95,666			0.081; OR 0.40 [0.19, 0.85]; n = 86		0.050; OR 0.26 [0.22, 0.31]; n = 2,662	
One-click tool	0.085; OR 0.38 [0.37, 0.38]; n = 352,569	0.069; OR 0.30 [0.29, 0.32]; n = 21,337	0.035; OR 0.14 [0.12, 0.16]; n = 4,806	0.066; OR 0.30 [0.29, 0.32]; n = 22,418	0.067; OR 0.31 [0.27, 0.34]; n = 4,280	0.053; OR 0.27 [0.26, 0.29]; n = 19,442	0.031; OR 0.09 [0.08, 0.10]; n = 7,052
Other registered, manual	0.113; OR 0.51 [0.51, 0.52]; n = 284,043	0.107; OR 0.49 [0.46, 0.51]; n = 17,054	0.119; OR 0.50 [0.45, 0.56]; n = 2,797	0.140; OR 0.70 [0.68, 0.72]; n = 32,379	0.132; OR 0.65 [0.61, 0.69]; n = 9,028	0.104; OR 0.57 [0.55, 0.59]; n = 51,753	0.125; OR 0.40 [0.38, 0.42]; n = 15,857
Unregistered	0.104; OR 0.47 [0.45, 0.49]; n = 17,863	0.126; OR 0.59 [0.49, 0.71]; n = 979	0.124; OR 0.53 [0.44, 0.65]; n = 916	0.087; OR 0.41 [0.33, 0.52]; n = 961	0.137; OR 0.68 [0.52, 0.90]; n = 438	0.081; OR 0.44 [0.35, 0.55]; n = 983	0.144; OR 0.47 [0.42, 0.53]; n = 2,275
Self	0.137; OR 0.64 [0.62, 0.66]; n = 34,890	0.126; OR 0.59 [0.52, 0.68]; n = 1,898	0.148; OR 0.65 [0.54, 0.77]; n = 960	0.114; OR 0.55 [0.48, 0.65]; n = 1,617	0.128; OR 0.63 [0.49, 0.81]; n = 538	0.119; OR 0.66 [0.60, 0.73]; n = 3,886	0.199; OR 0.69 [0.63, 0.75]; n = 3,423

Share retained among newcomers not reverted within 24 hours: English 0.198; Italian* 0.196; Korean 0.212; German 0.189; Polish* 0.190; French 0.169; Japanese* 0.265.

Share of newcomers blocked within 60 days of the first edit, by class of first revert within 24 hours.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
All newcomers	0.015	0.054	0.024	0.019	0.021	0.021	0.019
Not reverted	0.006	0.027	0.010	0.007	0.013	0.007	0.009
Bot	0.035			0.000		0.251	
One-click tool	0.028	0.121	0.093	0.099	0.096	0.096	0.083
Other registered, manual	0.035	0.094	0.059	0.020	0.025	0.046	0.087
Unregistered	0.024	0.134	0.073	0.046	0.035	0.058	0.085
Self	0.013	0.059	0.028	0.015	0.046	0.022	0.039

Table S6. Tag coverage: radius-rule reverts whose reverted first edit carries no mw-reverted tag, first edits from October 2020 (page-history samples)

Patrolling regime: English, Korean, Japanese none; German, Polish FlaggedRevs; Italian, French recent-changes patrolling. Cells: untagged share with 95 percent Wilson interval; number of reverts.

By class of reverter.

reverter_class	English	Italian*	Korean	German	Polish*	French	Japanese*
All reverters	0.003 [0.002, 0.004]; n = 16,611	0.096 [0.091, 0.101]; n = 14,935	0.004 [0.003, 0.005]; n = 9,485	0.024 [0.022, 0.027]; n = 12,246	0.039 [0.036, 0.043]; n = 13,282	0.070 [0.065, 0.075]; n = 10,914	0.010 [0.008, 0.013]; n = 7,039
Bot	0.000 [0.000, 0.003]; n = 2,012		0.000 [0.000, 0.071]; n = 50			0.004 [0.001, 0.024]; n = 231	
One-click tool	0.002 [0.001, 0.003]; n = 8,433	0.006 [0.005, 0.008]; n = 7,952	0.002 [0.001, 0.004]; n = 4,809	0.022 [0.018, 0.027]; n = 4,413	0.052 [0.045, 0.061]; n = 3,112	0.003 [0.002, 0.006]; n = 2,682	0.015 [0.010, 0.022]; n = 1,813
Other registered, manual	0.003 [0.002, 0.005]; n = 4,942	0.106 [0.099, 0.114]; n = 5,964	0.004 [0.002, 0.007]; n = 2,793	0.016 [0.013, 0.019]; n = 7,305	0.020 [0.017, 0.023]; n = 9,363	0.031 [0.027, 0.035]; n = 7,274	0.006 [0.004, 0.009]; n = 3,884
Unregistered	0.002 [0.000, 0.010]; n = 551	0.786 [0.745, 0.822]; n = 434	0.002 [0.000, 0.007]; n = 1,107	0.235 [0.179, 0.301]; n = 183	0.286 [0.245, 0.332]; n = 412	0.715 [0.653, 0.770]; n = 228	0.000 [0.000, 0.006]; n = 624
Self	0.022 [0.014, 0.036]; n = 673	0.708 [0.670, 0.744]; n = 579	0.022 [0.014, 0.035]; n = 726	0.127 [0.094, 0.170]; n = 299	0.135 [0.103, 0.174]; n = 364	0.733 [0.693, 0.770]; n = 499	0.027 [0.018, 0.042]; n = 697

By user group of the reverting account. The groups are those recorded for the account at the time of the revert (event_user_groups_historical). Whether a group carries the autopatrol right (recent-changes patrolling) or the autoreview right (FlaggedRevs) is taken from each edition's configuration as read in November 2026 (Endnote 1 of the article); the configuration history was not reconstructed, so the classification is a proxy for the rights an account held when it reverted, and patrol or review actions themselves were not observed. Implicit groups such as autoconfirmed, and the automatic French promotion, are not recorded in this field. The table with account-resampled intervals further below gives the cells.

Registered manual reverters other than the author with no explicit group, by the account's edit count at the time of the revert. On the French edition the autopatrolled status is granted automatically at 500 edits and 90 days and is not recorded as an explicit group. Account age at the time of the revert was not extracted, so the edit count is a proxy for that status, not a determination of it.

reverter_edit_count	English	Italian*	Korean	German	Polish*	French	Japanese*
under 50	0.010 [0.003, 0.036]; n = 199	0.718 [0.610, 0.806]; n = 78	0.000 [0.000, 0.014]; n = 262		0.377 [0.272, 0.495]; n = 69	0.701 [0.608, 0.779]; n = 107	0.000 [0.000, 0.015]; n = 249
50 to 499	0.003 [0.001, 0.017]; n = 330	0.690 [0.606, 0.763]; n = 129	0.000 [0.000, 0.020]; n = 187		0.289 [0.205, 0.390]; n = 90	0.597 [0.524, 0.665]; n = 181	0.003 [0.001, 0.016]; n = 353
500 to 4,999		0.826 [0.780, 0.865]; n = 305			0.111 [0.060, 0.198]; n = 81	0.034 [0.024, 0.049]; n = 789	0.005 [0.001, 0.027]; n = 205
5,000 or more		0.755 [0.696, 0.806]; n = 233			0.133 [0.084, 0.206]; n = 120	0.001 [0.000, 0.003]; n = 3,274	

All reverters, by year of the first edit.

year	English	Italian*	Korean	German	Polish*	French	Japanese*
2020	0.003	0.090	0.000	0.016	0.021	0.071	0.008
2021	0.003	0.092	0.006	0.023	0.023	0.073	0.009
2022	0.003	0.088	0.005	0.026	0.039	0.078	0.016
2023	0.002	0.088	0.002	0.019	0.041	0.062	0.010
2024	0.004	0.093	0.004	0.038	0.054	0.070	0.005
2025	0.002	0.121	0.005	0.020	0.048	0.060	0.006
2026	0.002	0.141	0.003	0.015	0.067	0.089	0.015

Worked example: one cross-edition comparison computed with the tag and with the radius rule (first edits from October 2020; share of first edits counted as reverted; gap to English in percentage points; odds ratio of retention for reverted against non-reverted newcomers).

Edition	n	Reverted, tag	Reverted, radius rule	Gap to English, tag	Gap to English, radius rule	Change in gap [95% interval, editors resampled]	[95% interval, reverting accounts resampled]	OR retention, tag	OR retention, radius rule
English	42,042	0.401	0.395	0	0			0.54	0.53
Italian*	41,609	0.327	0.359	7.4	3.6	3.8 [3.6, 4.0]	[3.4, 4.2]	0.46	0.45
Korean	34,456	0.28	0.275	12.1	12	0.1 [0.0, 0.3]	[-0.0, 0.3]	0.48	0.48
German	42,644	0.282	0.287	11.9	10.8	1.1 [1.0, 1.3]	[1.0, 1.3]	0.58	0.58
Polish*	42,613	0.303	0.312	9.9	8.3	1.5 [1.4, 1.7]	[1.3, 1.8]	0.61	0.61
French	42,312	0.243	0.258	15.8	13.7	2.1 [1.9, 2.2]	[1.9, 2.3]	0.61	0.6
Japanese*	42,442	0.171	0.166	23	22.9	0.1 [-0.1, 0.2]	[-0.1, 0.2]	0.46	0.46

Tag by radius rule, cell by cell (first edits from October 2020). The two rules are not nested: MediaWiki records an edit made through the undo function as a revert whether or not the result

is identical to an earlier revision, so the tag also marks edits that no hash-based rule finds. Dump flag: the dumps' own identity-revert field, computed by Wikimedia on the complete page history. The last column bounds the effect of loading the English history from 2018: edits that the dumps' flag marks reverted and the recomputation on the loaded history does not.

Edition	n	Both	Tag only	Radius rule only	Neither	Tag only, % of first edits	Radius rule only, % of first edits	Tag only: reverted under the dump flag	Tag only: not reverted under the dump flag	Dump flag yes, recomputation no
English	42,042	16,566	301	45	25,130	0.72	0.11	9	292	28
Italian*	41,609	13,502	108	1,433	26,566	0.26	3.44	3	105	0
Korean	34,456	9,448	199	37	24,772	0.58	0.11	17	182	1
German	42,644	11,950	66	296	30,332	0.15	0.69	1	65	0
Polish*	42,613	12,760	134	522	29,197	0.31	1.22	0	134	0
French	42,312	10,149	150	765	31,248	0.35	1.81	3	147	0
Japanese*	42,442	6,968	307	71	35,096	0.72	0.17	3	304	0

Decomposition of the change in the gap to English between the two rules, by period of the first edit (percentage points). Change = radius-only share in the edition, minus its tag-only share, plus the English tag-only share, minus the English radius-only share.

Edition	Period	Gap, tag	Gap, radius rule	Change	Radius only, edition	Tag only, edition (minus)	Tag only, English	Radius only, English (minus)
Italian*	from 2020-10	7.41	3.62	3.79	3.44	-0.26	0.72	-0.11
Korean	from 2020-10	12.12	11.98	0.14	0.11	-0.58	0.72	-0.11
German	from 2020-10	11.94	10.79	1.15	0.69	-0.15	0.72	-0.11
Polish*	from 2020-10	9.86	8.34	1.52	1.22	-0.31	0.72	-0.11
French	from 2020-10	15.78	13.72	2.06	1.81	-0.35	0.72	-0.11
Japanese*	from 2020-10	22.98	22.93	0.05	0.17	-0.72	0.72	-0.11
Italian*	2020-10 to 2024-12	6.86	3.12	3.74	3.36	-0.26	0.76	-0.12
Korean	2020-10 to 2024-12	12.96	12.81	0.15	0.10	-0.59	0.76	-0.12
German	2020-10 to 2024-12	11.98	10.74	1.24	0.75	-0.16	0.76	-0.12
Polish*	2020-10 to 2024-12	8.86	7.34	1.52	1.22	-0.34	0.76	-0.12
French	2020-10 to 2024-12	16.05	13.87	2.18	1.88	-0.34	0.76	-0.12
Japanese*	2020-10 to 2024-12	23.03	22.96	0.08	0.18	-0.75	0.76	-0.12
Italian*	from 2025-01	9.57	5.58	3.99	3.77	-0.25	0.53	-0.06
Korean	from 2025-01	8.56	8.46	0.10	0.13	-0.50	0.53	-0.06
German	from 2025-01	11.42	10.63	0.78	0.47	-0.15	0.53	-0.06
Polish*	from 2025-01	13.36	11.86	1.49	1.25	-0.22	0.53	-0.06
French	from 2025-01	14.72	13.16	1.56	1.49	-0.40	0.53	-0.06
Japanese*	from 2025-01	23.49	23.54	-0.04	0.09	-0.60	0.53	-0.06

Untagged share of radius-rule reverts by class of first reverter and period, editions with patrolling or FlaggedRevs. Classes with fewer than 50 reverts are blank.

reverter	French, 2020-10 to 2024-12	French, from 2025-01	German, 2020-10 to 2024-12	German, from 2025-01	Italian*, 2020-10 to 2024-12	Italian*, from 2025-01	Polish*, 2020-10 to 2024-12	Polish*, from 2025-01
all	0.071 (share of reverts 1.000)	0.066 (share of reverts 1.000)	0.025 (share of reverts 1.000)	0.018 (share of reverts 1.000)	0.090 (share of reverts 1.000)	0.125 (share of reverts 1.000)	0.037 (share of reverts 1.000)	0.052 (share of reverts 1.000)
tool	0.003 (share of reverts 0.257)	0.000 (share of reverts 0.187)	0.024 (share of reverts 0.363)	0.010 (share of reverts 0.348)	0.007 (share of reverts 0.538)	0.003 (share of reverts 0.502)	0.051 (share of reverts 0.238)	0.060 (share of reverts 0.214)
other manual	0.032 (share of reverts 0.654)	0.027 (share of reverts 0.730)	0.016 (share of reverts 0.593)	0.015 (share of reverts 0.614)	0.097 (share of reverts 0.394)	0.149 (share of reverts 0.425)	0.018 (share of reverts 0.705)	0.029 (share of reverts 0.704)
unregistered	0.753 (share of reverts 0.022)		0.233 (share of reverts 0.015)		0.759 (share of reverts 0.028)	0.888 (share of reverts 0.036)	0.291 (share of reverts 0.029)	0.270 (share of reverts 0.042)
self	0.744 (share of reverts 0.044)	0.691 (share of reverts 0.055)	0.131 (share of reverts 0.026)		0.699 (share of reverts 0.039)	0.761 (share of reverts 0.036)	0.119 (share of reverts 0.025)	0.190 (share of reverts 0.038)

Notes. The tag sits on the reverted edit and is set by any approved revert within its depth, including a later one by another account, while rows are classified by the first reverter under the radius rule; the class-specific untagged shares are therefore lower bounds on the omission per revert. Holders of global rights (global rollbackers, global sysops, stewards) have no local group in event_user_groups_historical and fall under “no group”. Manual patrol actions are recorded in the public logging table, which was not used; a join of reverting revisions to patrol log entries would test the mechanism directly.

Untagged share with intervals that resample reverting accounts (2,000 resamples; unregistered reverters and self-reverts are their own clusters). Here the group is one that carries the autopatrol right (Italian: autopatrolled, botadmin, rollbacker, sysop, bot; French: autopatrolled, sysop, bot) or the autoreview right (German, Polish: editor, reviewer, autoreview, sysop, bot) in the edition’s own configuration; in English, Korean and Japanese, where neither regime applies to ordinary pages, a common list of administrative and patrol groups is used. Intervals here and in Table S10 come from separate bootstrap runs and can differ in the third decimal.

Edition	Reverter	Reverts	Accounts	Share of the most active account	Untagged [Wilson]; [reverting accounts resampled]
English	all	16611	4849	0.114	0.003 [0.002, 0.004]; [0.002, 0.004]
English	bot	2012	9	0.939	0.000 [0.000, 0.003]; [0.000, 0.001]
English	tool	8433	1891	0.046	0.002 [0.001, 0.003]; [0.001, 0.002]
English	other manual	4942	2350	0.019	0.003 [0.002, 0.005]; [0.001, 0.005]
English	unregistered	551	551	0.002	0.002 [0.000, 0.010]; [0.000, 0.005]
English	self	673	673	0.001	0.022 [0.014, 0.036]; [0.012, 0.034]
English	tool, autopatrol-carrying group	6862	1298	0.057	0.001 [0.001, 0.002]; [0.001, 0.002]
English	tool, no such group	1571	698	0.049	0.003 [0.001, 0.007]; [0.001, 0.005]

Edition	Reverter	Reverts	Accounts	Share of the most active account	Untagged [Wilson]; [reverting accounts resampled]
English	other manual, autopatrol-carrying group	2397	919	0.024	0.004 [0.002, 0.007]; [0.001, 0.008]
English	other manual, no such group	2545	1498	0.038	0.002 [0.001, 0.005]; [0.001, 0.004]
Italian*	all	14935	1772	0.072	0.096 [0.091, 0.101]; [0.077, 0.123]
Italian*	tool	7952	166	0.082	0.006 [0.005, 0.008]; [0.004, 0.008]
Italian*	other manual	5964	727	0.071	0.106 [0.099, 0.114]; [0.084, 0.138]
Italian*	unregistered	434	434	0.002	0.786 [0.745, 0.822]; [0.749, 0.823]
Italian*	self	579	579	0.002	0.708 [0.670, 0.744]; [0.672, 0.746]
Italian*	tool, autopatrol-carrying group	7868	155	0.083	0.006 [0.005, 0.008]; [0.004, 0.008]
Italian*	tool, no such group	84	11	0.774	0.000 [0.000, 0.044]; [0.000, 0.000]
Italian*	other manual, autopatrol-carrying group	5212	429	0.081	0.010 [0.008, 0.013]; [0.006, 0.015]
Italian*	other manual, no such group	752	345	0.051	0.771 [0.740, 0.800]; [0.718, 0.819]
Korean	all	9485	2407	0.138	0.004 [0.003, 0.005]; [0.003, 0.005]
Korean	bot	50	5	0.440	0.000 [0.000, 0.071]; [0.000, 0.000]
Korean	tool	4809	75	0.226	0.002 [0.001, 0.004]; [0.001, 0.003]
Korean	other manual	2793	556	0.106	0.004 [0.002, 0.007]; [0.001, 0.006]
Korean	unregistered	1107	1107	0.001	0.002 [0.000, 0.007]; [0.000, 0.005]
Korean	self	726	726	0.001	0.022 [0.014, 0.035]; [0.012, 0.033]
Korean	tool, autopatrol-carrying group	4748	63	0.229	0.002 [0.001, 0.004]; [0.001, 0.003]
Korean	tool, no such group	61	12	0.328	0.000 [0.000, 0.059]; [0.000, 0.000]
Korean	other manual, autopatrol-carrying group	1432	80	0.207	0.007 [0.004, 0.013]; [0.003, 0.012]
Korean	other manual, no such group	1361	501	0.052	0.000 [0.000, 0.003]; [0.000, 0.000]
German	all	12246	2110	0.044	0.024 [0.022, 0.027]; [0.021, 0.028]
German	tool	4413	644	0.089	0.022 [0.018, 0.027]; [0.018, 0.027]
German	other manual	7305	1435	0.059	0.016 [0.013, 0.019]; [0.013, 0.019]
German	unregistered	183	183	0.005	0.235 [0.179, 0.301]; [0.175, 0.295]
German	self	299	299	0.003	0.127 [0.094, 0.170]; [0.090, 0.167]
German	tool, autopatrol-carrying group	4409	640	0.089	0.022 [0.018, 0.027]; [0.017, 0.027]
German	other manual, autopatrol-carrying group	7207	1385	0.059	0.013 [0.011, 0.016]; [0.011, 0.016]
German	other manual, no such group	98	64	0.153	0.204 [0.136, 0.294]; [0.113, 0.325]
Polish*	all	13282	1592	0.098	0.039 [0.036, 0.043];

Edition	Reverter	Reverts	Accounts	Share of the most active account	Untagged [Wilson]; [reverting accounts resampled]
					[0.031, 0.050]
Polish*	tool	3112	303	0.130	0.052 [0.045, 0.061]; [0.040, 0.064]
Polish*	other manual	9363	747	0.132	0.020 [0.017, 0.023]; [0.014, 0.027]
Polish*	unregistered	412	412	0.002	0.286 [0.245, 0.332]; [0.245, 0.333]
Polish*	self	364	364	0.003	0.135 [0.103, 0.174]; [0.102, 0.170]
Polish*	tool, autopatrol-carrying group	3056	298	0.133	0.053 [0.046, 0.062]; [0.040, 0.065]
Polish*	tool, no such group	56	7	0.875	0.018 [0.003, 0.094]; [0.000, 0.300]
Polish*	other manual, autopatrol-carrying group	9002	622	0.138	0.012 [0.010, 0.015]; [0.008, 0.017]
Polish*	other manual, no such group	361	146	0.219	0.216 [0.177, 0.261]; [0.135, 0.329]
French	all	10914	1990	0.043	0.070 [0.065, 0.075]; [0.058, 0.085]
French	bot	231	4	0.913	0.004 [0.001, 0.024]; [0.000, 0.005]
French	tool	2682	203	0.128	0.003 [0.002, 0.006]; [0.001, 0.007]
French	other manual	7274	1210	0.038	0.031 [0.027, 0.035]; [0.025, 0.038]
French	unregistered	228	228	0.004	0.715 [0.653, 0.770]; [0.658, 0.772]
French	self	499	499	0.002	0.733 [0.693, 0.770]; [0.693, 0.770]
French	tool, autopatrol-carrying group	1764	86	0.194	0.002 [0.001, 0.006]; [0.000, 0.005]
French	tool, no such group	918	124	0.112	0.004 [0.002, 0.011]; [0.000, 0.013]
French	other manual, autopatrol-carrying group	1389	95	0.107	0.006 [0.003, 0.012]; [0.003, 0.011]
French	other manual, no such group	5885	1130	0.047	0.037 [0.033, 0.042]; [0.029, 0.046]
Japanese*	all	7039	2441	0.073	0.010 [0.008, 0.013]; [0.007, 0.014]
Japanese*	tool	1813	70	0.237	0.015 [0.010, 0.022]; [0.007, 0.023]
Japanese*	other manual	3884	1099	0.062	0.006 [0.004, 0.009]; [0.003, 0.011]
Japanese*	unregistered	624	624	0.002	0.000 [0.000, 0.006]; [0.000, 0.000]
Japanese*	self	697	697	0.001	0.027 [0.018, 0.042]; [0.016, 0.040]
Japanese*	tool, autopatrol-carrying group	1773	62	0.228	0.015 [0.010, 0.022]; [0.007, 0.023]
Japanese*	other manual, autopatrol-carrying group	426	53	0.192	0.040 [0.025, 0.063]; [0.012, 0.094]
Japanese*	other manual, no such group	3458	1062	0.069	0.002 [0.001, 0.004]; [0.001, 0.004]

With first edits that create a page removed from the reference group (page-history samples, dump rule, exploratory). Odds ratios of editing again on days 2 to 60 against newcomers not reverted within 24 hours.

Edition	Scope	n	Retained, not reverted	OR tool	OR manual	OR self
English	all first edits	59,680	0.197	0.369	0.521	0.569
English	first edits to existing pages	59,670	0.197	0.370	0.521	0.570
Italian*	all first edits	59,999	0.202	0.306	0.463	0.539
Italian*	first edits to existing pages	50,682	0.223	0.271	0.409	0.476
Korean	all first edits	47,629	0.213	0.133	0.496	0.665
Korean	first edits to existing pages	40,727	0.213	0.134	0.497	0.667
German	all first edits	59,995	0.188	0.283	0.716	0.569
German	first edits to existing pages	50,397	0.198	0.265	0.671	0.533
Polish*	all first edits	59,997	0.191	0.300	0.666	0.627
Polish*	first edits to existing pages	47,965	0.212	0.265	0.587	0.552
French	all first edits	59,985	0.169	0.284	0.570	0.699
French	first edits to existing pages	52,708	0.182	0.258	0.518	0.635
Japanese*	all first edits	60,000	0.264	0.087	0.397	0.663
Japanese*	first edits to existing pages	54,410	0.263	0.088	0.400	0.669

Table S7. Registered tests in the held-out editions, as produced by the registered code

Bootstrap replicates in the registered code: 2,000 for H1 and H2, 500 for H4 (H4 intervals recomputed with 2,000 replicates are given below).

	Japanese*	Polish*	Italian*
H1: first edits in the sample	60,000	59,997	59,999
H1: replace effect within 24 hours (points)	2.972	6.217	15.900
H1: replace effect, unbounded (points)	2.730	5.450	13.409
H1: difference D (points)	0.242	0.767	2.492
H1: D, 90% lower bound	0.188	0.680	2.362
H1: D, one-sided 95% lower bound	0.180	0.660	2.338
H1: D, 95% upper bound	0.307	0.872	2.650
H1: ratio R	0.919	0.877	0.843
H1: R, 95% lower bound	0.898	0.861	0.834
H1: R, 95% upper bound	0.939	0.892	0.852
H2: first edits in the sample	60,000	59,997	59,999
H2: strict effect within 24 hours (points)	3.477	6.995	17.955
H2: replace effect within 24 hours (points)	2.972	6.217	15.900
H2: share S	0.145	0.111	0.114
H2: S, one-sided 95% upper bound	0.157	0.119	0.119
H2: S, 95% lower bound	0.131	0.102	0.109
H2: S, 95% upper bound	0.160	0.120	0.121
H3: first edits from 1 August 2020	44,211	44,041	43,006
H3: agreement, tag and radius rule	0.986	0.979	0.954
H3: agreement, tag and dump rule	0.981	0.977	0.951
H3: reverted under the dump rule, untagged (n)	538	876	1,997
H3: of these, share beyond the radius	0.424	0.094	0.073
H4: largest over smallest odds ratio, nine strict cells	1.836	1.425	1.783
H4: ratio, 95% lower bound	1.791	1.375	1.720
H4: ratio, 95% upper bound	1.878	1.494	1.841
H4: bootstrap replicates in the registered code	500	500	500

Output names in <wiki>_confirmatory.csv, in row order: n, delta24_pp, deltainf_pp, D_pp, D_ci90_lo, D_ci95_lo, D_ci95_hi, R, R_ci_lo, R_ci_hi, h2_n, h2_delta_strict_pp, h2_delta_replace_pp, h2_S, h2_S_ci95_hi_onesided, h2_S_ci_lo, h2_S_ci_hi, h3_n, h3_agree_d2_d3r15, h3_agree_d1_d2, h3_n_d1only, h3_d1only_beyond_radius, h4_or_ratio, h4_or_ratio_ci_lo, h4_or_ratio_ci_hi, h4_boot.

H4 recomputed with 2,000 paired bootstrap replicates, nine registered cells (ratio of the largest to the smallest odds ratio of retention across cells).

Edition	max/min odds ratio	95% interval	one-sided 95% lower bound
English	1.785	[1.768, 1.801]	1.771
Italian*	1.783	[1.722, 1.846]	1.732
Korean	3.008	[2.801, 3.221]	2.84
German	1.629	[1.592, 1.665]	1.599
Polish*	1.425	[1.368, 1.492]	1.379
French	1.512	[1.485, 1.538]	1.489
Japanese*	1.836	[1.791, 1.887]	1.798

H4 recomputed with 2,000 paired bootstrap replicates, three 24-hour cells (ratio of the largest to the smallest odds ratio of retention across cells).

Edition	max/min odds ratio	95% interval	one-sided 95% lower bound
English	1.492	[1.479, 1.504]	1.48
Italian*	1.503	[1.454, 1.552]	1.462
Korean	1.933	[1.835, 2.028]	1.851
German	1.488	[1.461, 1.516]	1.466
Polish*	1.302	[1.265, 1.344]	1.272
French	1.3	[1.283, 1.317]	1.285
Japanese*	1.321	[1.304, 1.339]	1.307

Table S8. Registered robustness analyses (bots and one-click tools excluded together, as registered)

Replace effect in percentage points by window (dump rule, page-history samples).

W	English	Italian*	Korean	German	Polish*	French	Japanese*
12 h	18.65	15.70	9.70	9.52	6.12	6.30	3.00
24 h	18.83	15.90	9.83	9.59	6.22	6.36	2.97
48 h	18.79	15.79	9.76	9.56	6.24	6.34	2.94
7 d	18.46	15.17	9.43	9.48	6.11	6.17	2.94
30 d	17.87	14.33	8.84	9.31	5.89	5.89	2.87
180 d	17.44	13.62	7.95	8.92	5.63	5.56	2.75
unbounded	17.37	13.41	7.62	8.79	5.45	5.45	2.73

Strict effect in percentage points by filter variant, 24 hours (full cohorts).

variant	English	Italian*	Korean	German	Polish*	French	Japanese*
F1 bots (current or historical)	4.53	0.01	0.07	0.04	0.03	1.02	0.01
F1c bots (current only)	4.53	0.01	0.03	0.04	0.02	1.02	0.01
F1fast bots + <=5 s registered	4.57	0.02	0.08	0.07	0.03	1.03	0.01
F1h bots (historical only)	4.53	0.01	0.07	0.04	0.03	1.02	0.01

variant	English	Italian*	Korean	German	Polish*	French	Japanese*
F2 bots + tools (rollback/huggle/twinkle)	21.09	17.94	11.19	10.54	7.01	7.22	3.46
F2fast bots + tools + <=5 s	21.10	17.94	11.20	10.54	7.01	7.23	3.46
F2s bots + tools + undo	32.33	28.11	17.06	17.85	14.56	18.91	9.15
F3 bots + tools + anonymous reverters	21.92	18.77	13.27	10.96	7.68	7.53	4.57

Bot-or-tool share of first reverts within 24 hours by cohort variant.

cohort	English	Italian*	Korean	German	Polish*	French	Japanese*
all registered first edits 2019+	0.565	0.523	0.511	0.398	0.317	0.292	0.259
self-created first edits 2019+ (primary)	0.570	0.525	0.519	0.411	0.316	0.297	0.248
self-created, anchors 2021+ (all tags available)	0.649	0.546	0.555	0.395	0.305	0.282	0.301
self-created, secondary anchors (edits 2-5 within 91 d)	0.564	0.517	0.512	0.409	0.335	0.315	0.280

Korean, with and without the surge of June 2023 to December 2024.

Cohort	First edits	p reverted, 24 h	Bot-or-tool share, 24 h	p reverted, unbounded	R, dump rule
all	47633	0.216	0.519	0.279	0.775
excl. 2023-06..2024-12 surge	35065	0.165	0.437	0.226	0.865
surge only	12568	0.357	0.624	0.427	0.692

Leave-one-edition-out and by-year results are in 04_output/robust_dose_response_loo.csv and 04_output/<edition>_grid_by_year.csv.

Table S9. Change tags on reverting revisions by edition and year (all namespaces)

mw-rollback

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1122426	895626	932095	883106	750250	774662	589221	606866	445436
Italian*	119804	124817	147566	159560	156810	127405	103429	90733	44813
Korean	20961	14881	28911	22528	31437	27802	29237	28939	11935
German	193873	189579	154538	142501	117173	120366	97322	69006	44425
Polish*	33863	33984	42665	37281	36801	33087	34604	22962	12179
French	82033	86959	68743	76641	67687	60521	59402	47767	33146
Japanese*	9121	11759	18030	27752	24273	26174	19051	14461	8951

huggle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	345176	262033	162423	1631	2409	151939	132999	104257	47488
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	5016	4433	836	1111	1774	4027	1868	1510	280
Japanese*	0	0	0	0	0	0	0	0	0

twinkle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	0	0	207362	602187	543831	514898	541969	538128	322779
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	0	0	0	0	0	0	0	0	0
Japanese*	0	0	0	0	0	0	0	0	0

mw-undo

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1494822	1529089	1540633	1552495	1435631	1340784	1362885	1285981	803483
Italian*	95578	117098	116429	114948	97846	89979	84370	73064	45823
Korean	30946	29764	57931	45761	26982	21147	28007	21249	11311
German	109547	96185	98655	97514	81234	79920	75943	69172	44466
Polish*	24450	26055	33152	46991	49128	31284	32403	22045	13004
French	161866	154328	168055	151144	134478	137671	119977	118490	84144
Japanese*	74816	95772	95019	93695	94968	67232	50025	44876	31462

Table S10. Sensitivity of intervals to clustering in reverting accounts, and H3 under two windows (page-history samples)

Editor-resampled intervals here and in Table S4 come from separate bootstrap runs and can differ in the third decimal. Coverage of account resampling: done for H1 (D and R under both rules), H2, the odds ratio of retention for tool-reverted against non-reverted newcomers, the untagged share (this table and Table S6) and the worked example (Table S6); not done for H4, for the odds ratios of manual and self-reverts, or for the direct contrast between manual and tool reverts, which the article reports as a descriptive range over complete cohorts.

Cells: estimate; 95 percent percentile interval resampling editors; 95 percent percentile interval resampling the accounts that made the first revert (non-reverted edits and edits first reverted by an unregistered editor are their own clusters; 1,000 replicates). No registered decision changes: under account resampling the one-sided 95 percent lower bound of D is above 0 and the upper bound of S below .25 in the three held-out editions.

quantity	English	Italian*	Korean	German	Polish*	French	Japanese*
H1 difference D, dump rule (points)	1.463; editors [1.312, 1.610]; accounts [0.753, 2.405]	2.492; editors [2.343, 2.638]; accounts [1.875, 3.086]	2.213; editors [2.030, 2.400]; accounts [0.954, 3.587]	0.793; editors [0.695, 0.890]; accounts [0.476, 1.122]	0.767; editors [0.657, 0.873]; accounts [0.431, 1.115]	0.905; editors [0.815, 1.015]; accounts [0.543, 1.341]	0.242; editors [0.177, 0.305]; accounts [0.051, 0.469]
R, dump rule	0.922; editors [0.915, 0.930]; accounts [0.901, 0.949]	0.843; editors [0.835, 0.852]; accounts [0.828, 0.859]	0.775; editors [0.758, 0.791]; accounts [0.720, 0.838]	0.917; editors [0.907, 0.927]; accounts [0.901, 0.939]	0.877; editors [0.861, 0.894]; accounts [0.854, 0.911]	0.858; editors [0.842, 0.872]; accounts [0.837, 0.885]	0.919; editors [0.899, 0.939]; accounts [0.886, 0.974]
R, radius rule	1.045; editors [1.040, 1.050]; accounts [1.028, 1.067]	1.024; editors [1.020, 1.029]; accounts [1.016, 1.034]	1.074; editors [1.063, 1.085]; accounts [1.050, 1.111]	1.036; editors [1.031, 1.042]; accounts [1.022, 1.055]	1.067; editors [1.057, 1.077]; accounts [1.044, 1.099]	1.039; editors [1.031, 1.046]; accounts [1.023, 1.062]	1.048; editors [1.037, 1.060]; accounts [1.023, 1.098]
H2 share S	0.100; editors	0.114; editors	0.122; editors	0.089; editors	0.111; editors	0.120; editors	0.145; editors

quantity	English	Italian*	Korean	German	Polish*	French	Japanese*
(registered definition)	[0.095, 0.105]; accounts [0.083, 0.116]	[0.109, 0.120]; accounts [0.103, 0.124]	[0.113, 0.130]; accounts [0.104, 0.141]	[0.082, 0.096]; accounts [0.080, 0.098]	[0.102, 0.121]; accounts [0.101, 0.124]	[0.110, 0.129]; accounts [0.090, 0.154]	[0.129, 0.160]; accounts [0.128, 0.173]
Odds ratio of retention, tool-reverted vs not reverted	0.370; editors [0.343, 0.398]; accounts [0.342, 0.399]	0.306; editors [0.283, 0.331]; accounts [0.261, 0.353]	0.134; editors [0.115, 0.155]; accounts [0.095, 0.201]	0.283; editors [0.252, 0.314]; accounts [0.244, 0.331]	0.301; editors [0.265, 0.336]; accounts [0.265, 0.345]	0.284; editors [0.244, 0.328]; accounts [0.241, 0.327]	0.088; editors [0.067, 0.109]; accounts [0.061, 0.114]
Untagged share of radius-rule reverts	0.003; editors [0.002, 0.004]; accounts [0.002, 0.004]	0.096; editors [0.091, 0.101]; accounts [0.076, 0.123]	0.004; editors [0.003, 0.005]; accounts [0.003, 0.005]	0.024; editors [0.021, 0.027]; accounts [0.021, 0.028]	0.039; editors [0.036, 0.043]; accounts [0.031, 0.049]	0.070; editors [0.065, 0.074]; accounts [0.059, 0.084]	0.010; editors [0.008, 0.012]; accounts [0.007, 0.014]

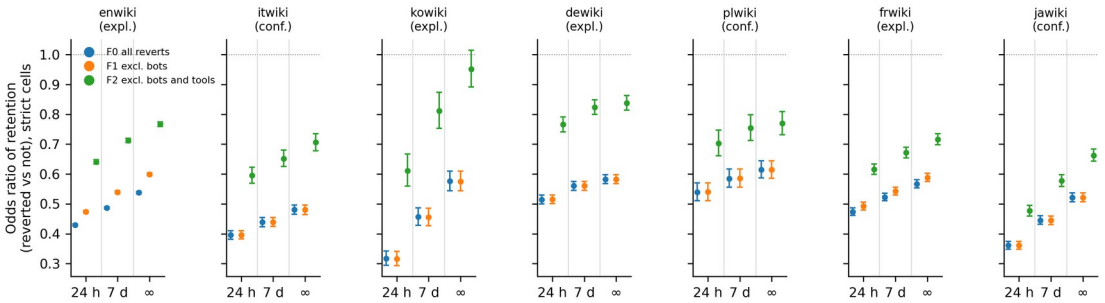
H3, second clause: of first edits reverted under the dump rule but not tagged, the share whose first revert lies beyond the 15-revision radius.

window	English	Italian*	Korean	German	Polish*	French	Japanese*
from 2020-10-01	0.841	0.088	0.954	0.137	0.131	0.114	0.749
registered window (from 2020-08-01)	0.340	0.073	0.770	0.075	0.094	0.096	0.424

Table S11. The proposed reporting items (Table 4 of the article) completed for this study

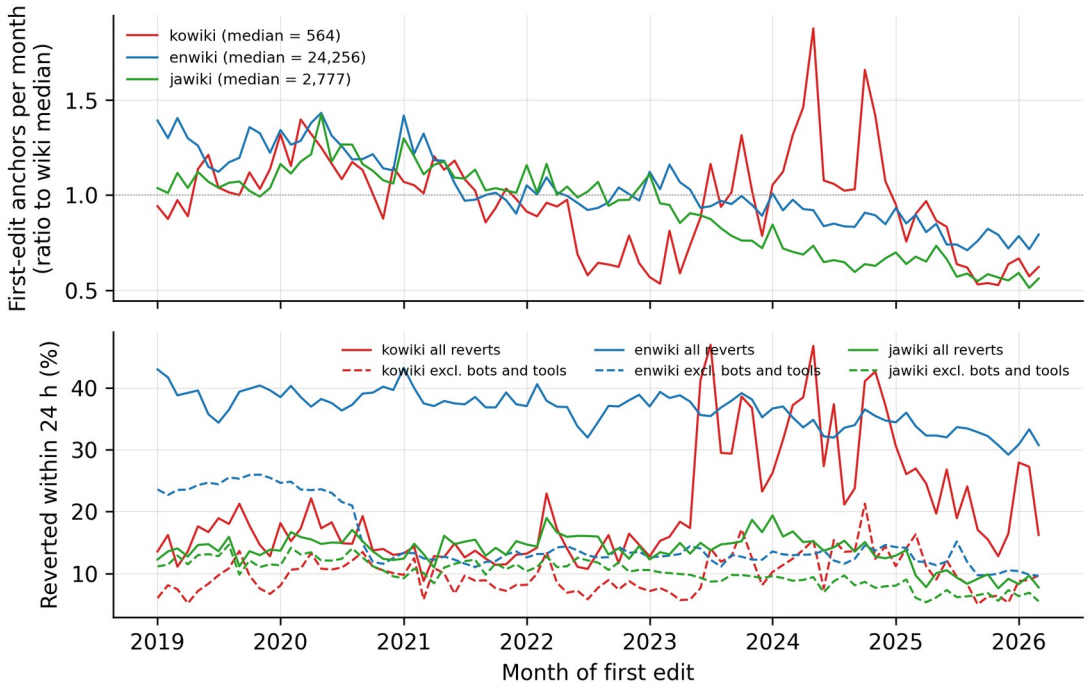
Item	This study
1. Construct	The audit is not tied to one construct; it reports what the indicator contains under each of four (Table 1). The cohort of newcomer first edits and the retention outcome correspond to the contributor's experience of rejection.
2. Attribution rule	Three, compared: the revision_is_identity_reverted field of the history dumps (unbounded); the same hashes with a 15-revision radius; the mw-reverted tag.
3. Window	24 hours, 7 days, 180 days and unbounded, all reported; 12 hours, 48 hours and 30 days in Table S8. No window is treated as validated for the non-English editions.
4. Bots, tools, author	Each counted and reported as a separate class; exclusions shown as variants (Table 3, Table S3).
5. Shares and accounts	Table 3 and Table S2, with the number of accounts making tool reverts and the share of the most active.
6. Outcomes by class	Retention by class of first reverter (Figure 3, Table S5).
7. Both rules	Follow-up and window results under the unbounded and the radius rule (Tables S4 and S10).
8. Tag	Patrolling regime and autopatrol rules per edition (Endnote 1 of the article); tag used for first edits from October 2020; coverage against the radius rule (Table S6).

Figure S1. Odds ratio of retention in the nine registered cells, by edition



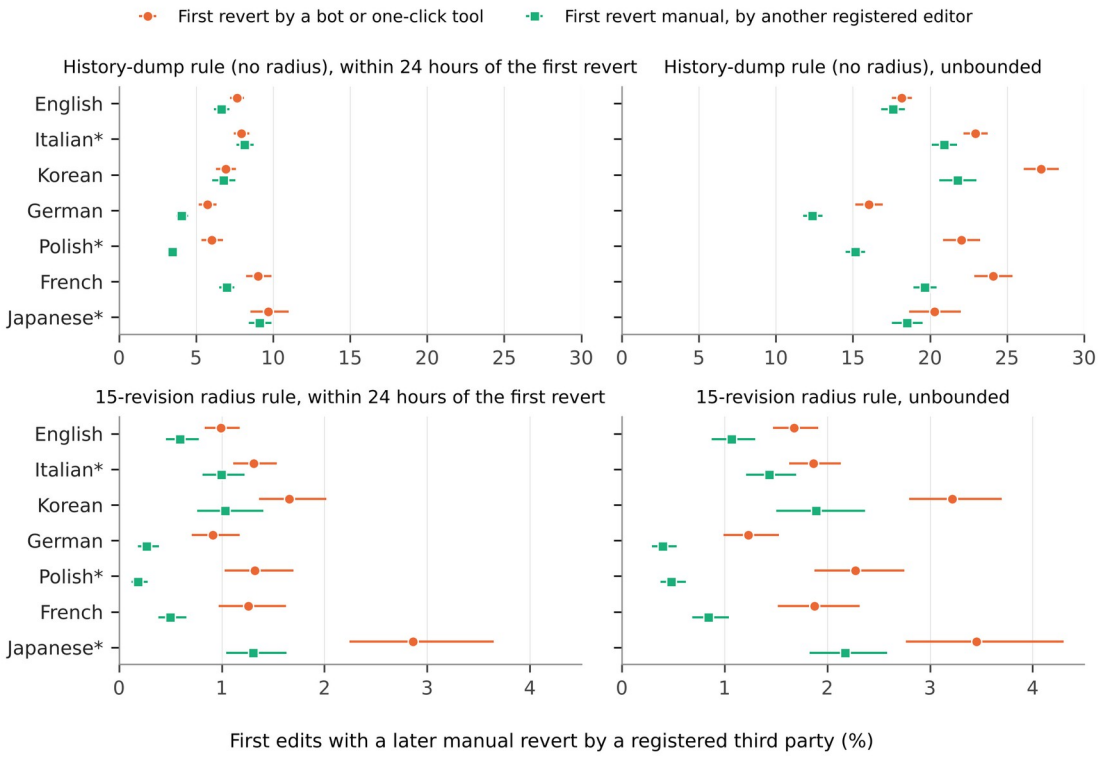
Reverted versus non-reverted first edits under three reverter filters (all reverts; without bots; without bots and one-click tools, strict recoding) and three windows. The movement across cells is compositional (see the article).

Figure S2. Monthly newcomer first edits and 24-hour revert rates in Korean, English and Japanese, 2019 to March 2026



Upper panel: first edits per month relative to the edition's median. Lower panel: share reverted within 24 hours, all reverts (solid) and without bots and one-click tools (dashed). The drop of the dashed English line in September 2020 marks the start of Twinkle tagging.

Figure S3. First edits with a later manual revert by a registered third party, by who made the first revert



Dump rule (top) and 15-revision radius rule (bottom), within 24 hours of the first revert (left) and unbounded (right). Bars are 95 percent Wilson intervals, which ignore clustering in reverting accounts. Exploratory.

Table S12. Page-history samples and validation of the attribution of first reverts

Sample flow. Eligible: first edits of self-created accounts, 1 January 2019 to 31 March 2026 (the primary cohort). Sampled: stratified by calendar year of the first edit with proportional allocation (the same sampling fraction in every year; all eligible edits in Korean), seed 20261026, so unweighted sample shares estimate cohort shares. Usable: the measured edit is found in the loaded page history and, for an edit to an existing page, so is the revision preceding it; first edits that create a page are retained (their shares are in Table S2). Tag analyses use the usable first edits from 1 October 2020 to 31 March 2026. Year-level counts are in 04_output/rev_sample_flow.csv.

Edition	Eligible	Sampled	Sampling fraction	Usable	Usable, from 1 Oct 2020	Of these, radius-reverted
English	2,189,284	60,000	0.0274	59,680	42,042	16,611
Italian*	135,400	60,000	0.4431	59,999	41,609	14,935

Edition	Eligible	Sampled	Sampling fraction	Usable	Usable, from 1 Oct 2020	Of these, radius-reverted
Korean	47,633	47,633	1	47,629	34,456	9,485
German	236,903	60,000	0.2533	59,995	42,644	12,246
Polish*	67,755	59,999	0.8855	59,997	42,613	13,282
French	346,938	60,001	0.1729	59,985	42,312	10,914
Japanese*	223,045	60,001	0.269	60,000	42,442	7,039

Which analysis uses which data.

Analysis	Data	First edits	Revert window	Other window
Composition of first reverts (Table 3, Figure 1, Table S2)	Full cohort	1 January 2019 to 31 March 2026	24 hours (7 days and unbounded in Table S3)	
Retention by class (Figure 3, Table S5)	Full cohort	same	24 hours	Activity on days 2 to 60; blocks within 60 days
Tag omission, worked example (Figure 2, Table S6)	Page-history sample	1 October 2020 to 31 March 2026	Unbounded to the dump (September 2026)	
Follow-up reverts, R, H1, H2 (Tables S4, S7, S10)	Page-history sample	1 January 2019 to 31 March 2026	24 hours and unbounded	
H3 (Table S7)	Page-history sample	1 August 2020 to 31 March 2026 as registered	Unbounded	
H4 (Table S7, Figure S1)	Full cohort	1 January 2019 to 31 March 2026	24 hours, 7 days, unbounded	Activity on days 2 to 60

Dump fields against the independent recomputation (dump rule, usable samples). Status: revision_is_identity_reverted against the recomputation from content hashes. Among first edits reverted under both, the first reverting revision and the delay to it are compared, and the class of the first reverter derived from each.

Status.

Edition	n	Reverted under both	Flag only	Recomputation only	Neither	Status agreement (%)
English	59,680	24,350	63	0	35,267	99.894
Italian*	59,999	22,096	0	0	37,903	100.000
Korean	47,629	13,288	9	0	34,332	99.981
German	59,995	17,682	1	0	42,312	99.998
Polish*	59,997	18,092	0	0	41,905	100.000
French	59,985	16,157	0	0	43,828	100.000
Japanese*	60,000	10,337	0	0	49,663	100.000

First revert, among first edits reverted under both. Percentages are rounded to three decimals, so a value of 100.000 can stand beside a non-zero count; the counts are exact.

Edition	Reverted under both	Same first reverting revision (%)	Different revision (n)	Crossing 24 hours (n)	Different class (n)
English	24,350	99.984	4	0	1
Italian*	22,096	99.995	1	0	0

Edition	Reverted under both	Same first reverting revision (%)	Different revision (n)	Crossing 24 hours (n)	Different class (n)
Korean	13,288	99.992	1	0	0
German	17,682	100.000	0	0	0
Polish*	18,092	100.000	0	0	0
French	16,157	99.994	1	0	1
Japanese*	10,337	100.000	0	0	0

Flag-only cases in English are reverts to states older than the loaded history (from 2018); elsewhere there are at most nine cases. No case moves across the 24-hour boundary, and the class of the first reverter differs in at most one case per edition.

Radius rule against python-mwreverts (version 0.1.5, Detector with radius 15) on synthetic page histories. For every revision, the first reverting revision under the implementation used here is compared with the first revert in which mwreverts lists the revision as reverted.

Synthetic histories	Pages	Revisions checked	Reverted under mwreverts	Mismatches
random, few states	400	13,152	12,022	0
random, many states	400	12,814	9,625	0
edit war (two states alternating with noise)	400	12,724	11,577	0
consecutive identical hashes	400	12,616	8,580	0
restoration at the 15-revision boundary	400	8,991	2,796	0
Total	2,000	60,297	44,600	0

Not covered by this validation: partial reverts, which no hash-based rule detects, and the tag rule, which is taken from the dumps as recorded.